

FADDIS Experiences with Real-World Networks

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Pipeline Thresholds

LAPIN-on Median Normalized Contributions (thresholds\lapin-on-median-normalized-contributions folder)

- 1) Stage 1: Obtain candidate thresholds for each network family
 - a) For each network in each network family, execute FADDIS using the default input matrix (adjacency matrix), with the LAPIN transformation as a pre-processing step, until it extracts, if possible, the ground-truth number of clusters, K;
 - b) Register the raw contribution values in the order in which they are extracted (raw_contributions.csv files);
 - c) Normalize the contributions by dividing each contribution value by the sum of all contributions from all networks across all families (normalized_contributions.csv files);
 - d) Compute statistical metrics for each network family based on the normalized contributions, namely the mean, sample standard deviation, median, 75th percentile, 90th percentile, 95th percentile, minimum and maximum (statistics.csv file);
 - e) Consider these statistical values as candidate thresholds (candidate_thresholds.csv file).
- 2) Stage 2: Choose one of the candidate thresholds for each network family
 - a) Due to the reduced number of networks compared with the experiences using LFR networks, instead of using bootstrap resampling to select the statistical metric that shows the greatest stability and robustness under resampling, use the previous knowledge obtained from those experiences and consider also the median for real-world networks (thresholds.csv file).
- 3) Stage 3: Perform a network-level sensitivity analysis
 - a) Evaluate the correlation between the network-level median normalized-contributions and the computed network properties (network_properties_train_set.csv file) for each network. These properties include both structural properties and properties related to the ground-truth community structure. The correlation is evaluated using both Spearman and Pearson correlation coefficients (faddis_sensitivity_correlations.csv file).

LAPIN-on Median K Boundary Raw Contributions (thresholds\lapin-on-median-k-boundary-raw-contributions folder)

- 1) Stage 1: Obtain K-boundary thresholds for each network
 - a) For each network in each network family, execute FADDIS using the default input matrix (adjacency matrix), with the LAPIN transformation as a pre-processing step, until the extraction process finishes;
 - b) Register the raw contribution values in the order in which they are extracted (raw_contributions.csv files);
 - c) Normalize the contributions by dividing each contribution value by the sum of all contributions from all networks across all families (normalized_contributions.csv files);
 - d) For each network, identify the normalized contribution at the ground-truth number of clusters, K , and the next normalized contribution, $K + 1$. Then, compute the network-level K -boundary threshold as the geometric mean between the normalized contribution at K and the normalized contribution at $K + 1$ (k_boundary_thresholds_by_network.csv file);
 - e) Consider the threshold valid only when the normalized contribution at $K + 1$ is smaller than the normalized contribution at K ;
 - f) Convert each valid normalized K -boundary threshold back to the raw contribution scale by multiplying it by the global normalization factor.
- 2) Stage 2: Choose the final threshold for each network family
 - a) For each network family, select the final threshold as the median of the valid raw K -boundary thresholds computed for the networks in that family (thresholds.csv file).
- 3) Stage 3: Perform a network-level sensitivity analysis
 - a) Evaluate the correlation between the network-level normalized contribution at K and the computed network properties (network_properties_train_set.csv file) for each network. These properties include both structural properties and properties related to the ground-truth community structure. The correlation is evaluated using both Spearman and Pearson correlation coefficients (faddis_sensitivity_correlations.csv file).

Real-World Networks

Network Selection and Division

- Networks pre-processed into **undirected, unweighted simple graphs without self-loops** and saved as *.gml* files.

Network	Knowledge Domain	Ground -Truth?	Nodes	Edges	CC	Nodes LCC	Edges LCC	Relative Size LCC	Overlapping Ground-Truth?	K LCC	Set
Zachary Karate Club	Human Social	Yes	34	78	1	34	78	1.0000	No	2	Train
Books about US Politics	Co-purchase	Yes	105	441	1	105	441	1.0000	No	3	Train
American College Football	Sports	Yes	115	613	1	115	613	1.0000	No	12	Train
SocioPatterns Primary School day 1	Human Contact	Yes	236	5899	1	236	5899	1.0000	No	11	Train
E-mail EU Core	Communication	Yes	1005	16064	20	986	16064	0.9811	No	42	Train
US Political Blogs	Web / Hyperlink	Yes	1490	16715	268	1222	16714	0.8201	No	2	Train
Cora	Scientific Citation	Yes	2708	5278	78	2485	5069	0.9177	No	7	Train
Facebook Ego-698 Network	Online Social	Yes	61	270	3	40	220	0.6557	Yes	9	Train
Facebook Ego-414 Network	Online Social	Yes	150	1693	2	148	1692	0.9867	Yes	7	Train
Facebook Ego-0 Network	Online Social	Yes	333	2519	5	324	2514	0.9730	Yes	22	Train
Facebook Ego-3437 Network	Online Social	Yes	534	4813	2	532	4812	0.9963	Yes	32	Train
Facebook Ego-1684 Network	Online Social	Yes	786	14024	4	775	14006	0.9860	Yes	17	Train
Facebook Ego-1912 Network	Online Social	Yes	747	30025	2	744	30023	0.9960	Yes	45	Train
Facebook Ego-107 Network	Online Social	Yes	1034	26749	1	1034	26749	1.0000	Yes	9	Train
Word Adjacencies	Language	Yes	112	425	1	112	425	1.0000	No	2	Test
SocioPatterns Primary School day 2	Human Contact	Yes	238	5539	1	238	5539	1.0000	No	11	Test
CiteSeer	Scientific Citation	Yes	3312	4536	438	2110	3668	0.6371	No	6	Test
Facebook Ego-3980 Network	Online Social	Yes	52	146	4	44	138	0.8462	Yes	11	Test
Facebook Ego-686 Network	Online Social	Yes	168	1656	1	168	1656	1.0000	Yes	14	Test

Network	Knowledge Domain	Ground-Truth?	Nodes	Edges	CC	Nodes LCC	Edges LCC	Relative Size LCC	Overlapping Ground-Truth?	K LCC	Set
Facebook Ego-348 Network	Online Social	Yes	224	3192	1	224	3192	1.0000	Yes	14	Test
Dolphins	Geographic	No	62	159	1	62	159	1.0000	-	-	Test
Les Miserables	Character	No	77	254	1	77	254	1.0000	-	-	Test
Jazz Musicians	Human Social	No	198	2742	1	198	2742	1.0000	-	-	Test
C.Elegans Neural Network	Neural	No	297	2148	1	297	2148	1.0000	-	-	Test
C.Elegans Metabolic	Metabolic	No	453	2025	1	453	2025	1.0000	-	-	Test
E-mail URV	Communication	No	1133	5451	1	1133	5451	1.0000	-	-	Test
Co-authorships in Network Science	Co-authorship	No	1589	2742	396	379	914	0.2385	-	-	Test

Table 1- Real-world networks

Networks Grouped by Degree Assortativity and Average Degree

Range Definitions

Degree Assortativity (Primary)

Category	Range
Disassortative	$r < -0.10$
Near-Neutral	$-0.10 \leq r < 0.10$
Assortative	$r \geq 0.10$

Table 2 - Degree assortativity ranges

Average Degree (Secondary)

Category	Range
Low Average Degree	$a < 15$
Medium Average Degree	$15 \leq a < 35$
Large Average Degree	$35 \leq a < 65$
Very Large Average Degree	$a \geq 65$

Table 3 - Average degree ranges

Network Families

Combination	Name
Disassortative + Low Average Degree	Network Family 1
Disassortative + Medium Average Degree	Network Family 2
Disassortative + Large Average Degree	-
Disassortative + Very Large Average Degree	-
Near-Neutral + Low Average Degree	Network Family 3
Near-Neutral + Medium Average Degree	Network Family 4
Near-Neutral + Large Average Degree	-
Near-Neutral + Very Large Average Degree	-
Assortative + Low Average Degree	Network Family 5
Assortative + Medium Average Degree	Network Family 6
Assortative + Large Average Degree	Network Family 7
Assortative + Very Large Average Degree	Network Family 8

Table 4 - Network families

Network Family 1 (Disassortative + Low Average Degree)

- Range
 - Disassortative: $r < -0.10$
 - Low Average Degree: $a < 15$

Network	Set	Average Degree	Degree Assortativity
zachary-karate-club	Train	4.5882	-0.4756
books-about-us-politics	Train	8.4000	-0.1279

Table 5 - Network Family 1 (Disassortative + Low Average Degree)

Network Family 2 (Disassortative + Medium Average Degree)

- Range
 - Disassortative: $r < -0.10$
 - Medium Average Degree: $15 \leq a < 35$

Network	Set	Average Degree	Degree Assortativity
us-political-blogs	Train	27.3552	-0.2213

Table 6 - Network Family 2 (Disassortative + Medium Average Degree)

Network Family 3 (Near-Neutral + Low Average Degree)

- Range
 - Near-Neutral: $-0.10 \leq r < 0.10$
 - Low Average Degree: $a < 15$

Network	Set	Average Degree	Degree Assortativity
cora	Train	4.0797	-0.0714
facebook-network-ego698	Train	11.0000	0.0125

Table 7 - Network Family 3 (Near-Neutral + Low Average Degree)

Network Family 4 (Near-Neutral + Medium Average Degree)

- Range
 - Near-Neutral: $-0.10 \leq r < 0.10$
 - Medium Average Degree: $15 \leq a < 35$

Network	Set	Average Degree	Degree Assortativity
email-eu-core	Train	32.5842	-0.0257

Table 8 - Network Family 4 (Near-Neutral + Medium Average Degree)

Network Family 5 (Assortative + Low Average Degree)

- Range
 - Assortative: $r \geq 0.10$
 - Low Average Degree: $a < 15$

Network	Set	Average Degree	Degree Assortativity
american-college-football	Train	10.6609	0.1624

Table 9 - Network Family 5 (Assortative + Low Average Degree)

Network Family 6 (Assortative + Medium Average Degree)

- Range
 - Assortative: $r \geq 0.10$
 - Medium Average Degree: $15 \leq a < 35$

Network	Set	Average Degree	Degree Assortativity
facebook-network-ego0	Train	15.5185	0.2330
facebook-network-ego3437	Train	18.0902	0.2221
facebook-network-ego414	Train	22.8649	0.3039

Table 10 - Network Family 6 (Assortative + Medium Average Degree)

Network Family 7 (Assortative + Large Average Degree)

- Range
 - Assortative: $r \geq 0.10$
 - Large Average Degree: $35 \leq a < 65$

Network	Set	Average Degree	Degree Assortativity
facebook-network-ego1684	Train	36.1445	0.3268
socio-patterns-primary-school-day1	Train	49.9915	0.1729
facebook-network-ego107	Train	51.7389	0.4316

Table 11 - Network Family 7 (Assortative + Large Average Degree)

Network Family 8 (Assortative + Very Large Average Degree)

- Range
 - Assortative: $r \geq 0.10$
 - Very Large Average Degree: $a \geq 65$

Network	Set	Average Degree	Degree Assortativity
facebook-network-ego1912	Train	80.7070	0.5026

Table 12 - Network Family 8 (Assortative + Very Large Average Degree)

Test Networks Assignment

Network properties ([thresholds\network_properties\network_properties_test_set.csv](#)
file)

Network	Average Degree	Degree Assortativity	Combination	Assigned Family	Calibrated?
word-adjacencies	7.5893	-0.1293	Disassortative + Low Average Degree	network_family_01	Yes
socio-patterns-primary-school-day2	46.5462	0.2168	Assortative + Large Average Degree	network_family_07	Yes
citeseer	3.4768	0.0071	Near-Neutral + Low Average Degree	network_family_03	Yes
facebook-network-ego3980	6.2727	0.0530	Near-Neutral + Low Average Degree	network_family_03	Yes
facebook-network-ego686	19.7143	0.0841	Near-Neutral + Medium Average Degree	network_family_04	Yes
facebook-network-ego348	28.5000	0.2227	Assortative + Medium Average Degree	network_family_06	Yes
dolphins	5.1290	-0.0436	Near-Neutral + Low Average Degree	network_family_03	Yes
les-miserables	6.5974	-0.1652	Disassortative + Low Average Degree	network_family_01	Yes
jazz-musicians	27.6970	0.0202	Near-Neutral + Medium Average Degree	network_family_04	Yes
c-elegans-neural-network	14.4646	-0.1632	Disassortative + Low Average Degree	network_family_01	Yes
c-elegans-metabolic	8.9404	-0.2258	Disassortative + Low Average Degree	network_family_01	Yes
email-urv	9.6222	0.0782	Near-Neutral + Low Average Degree	network_family_03	Yes
co-authorships-in-network-science	4.8232	-0.0817	Near-Neutral + Low Average Degree	network_family_03	Yes

Table 13 - Test networks assignment

Pipeline Results

Experiences

- Reference (results\reference\results.csv file);
- LAPIN-on Median Normalized Contributions (results\lapin-on-median-normalized-contributions\results.csv file);
- LAPIN-on Median K Boundary Raw Contributions (results\lapin-on-median-k-boundary-raw-contributions\results.csv file).

Pipeline

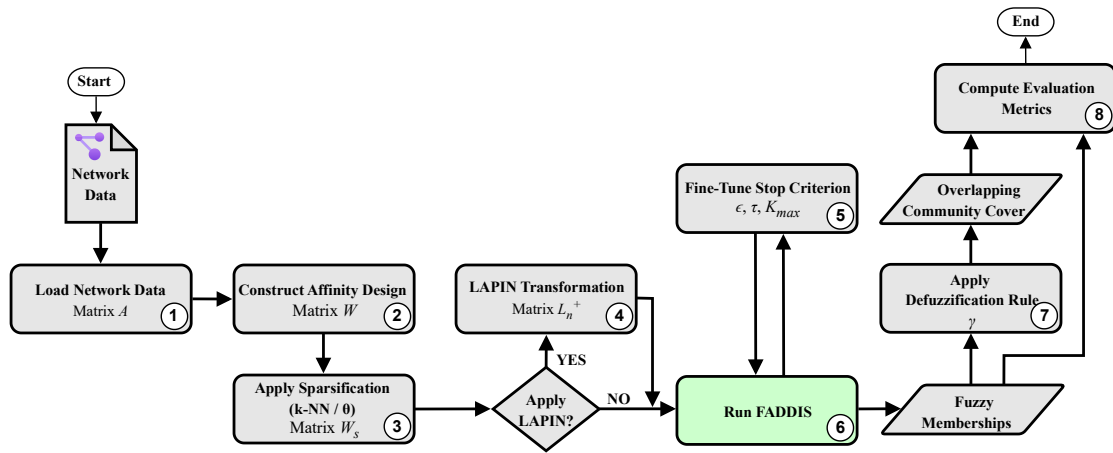


Figure 1 - Experimental pipeline

1. Select one real-world network from the experimental set;
2. Consider only the default affinity matrix, i.e., the adjacency matrix;
3. Do not apply sparsification;
4. Consider both execution modes: LAPIN-off and LAPIN-on;
5. Use the following stopping criterion:
 - a. In the *reference* experience, use the ground-truth number of communities, K , as the stopping criterion;
 - b. In the *LAPIN-on Median Normalized Contributions* and *LAPIN-on Median K-Boundary Raw Contributions* experiences, use ***epsilon***, defined by the corresponding network-family threshold, as the stopping criterion. And set ***tau*** = 0.05 and **$K_{max} = \min(500, \frac{n}{2})$** ;

6. Execute FADDIS using the parameters defined in the pipeline;
7. Apply the defuzzification rule according to the ground-truth type:
 - a. For networks with non-overlapping ground-truth, apply maximum-membership assignment;
 - b. For networks with overlapping ground-truth, apply node-wise α -cut thresholding relative to the maximum membership value, using ***gamma*** as a hyperparameter. The considered ***gamma*** values are: 0.3, 0.5, 0.6, 0.7, 0.8 and 0.9;
 - c. For networks without ground-truth, consider both defuzzification rules;
8. Select the evaluation metrics according to the ground-truth type:
 - a. For networks with non-overlapping ground-truth:
 - *Extrinsic metrics*: AMI as the primary metric; F-measure, ARI, FMI, NMI and VI as secondary metrics;
 - *Intrinsic metrics*: Modularity as the primary metric and Conductance as the secondary metric;
 - b. For networks with overlapping ground-truth:
 - *Extrinsic metrics*: ONMI as the primary metric and Omega as the secondary metric;
 - *Intrinsic metrics*: Fuzzy Modularity as the primary metric and Conductance of Boundary Nodes as the secondary metric;
 - c. For networks without ground-truth, evaluate the results using all considered intrinsic metrics.